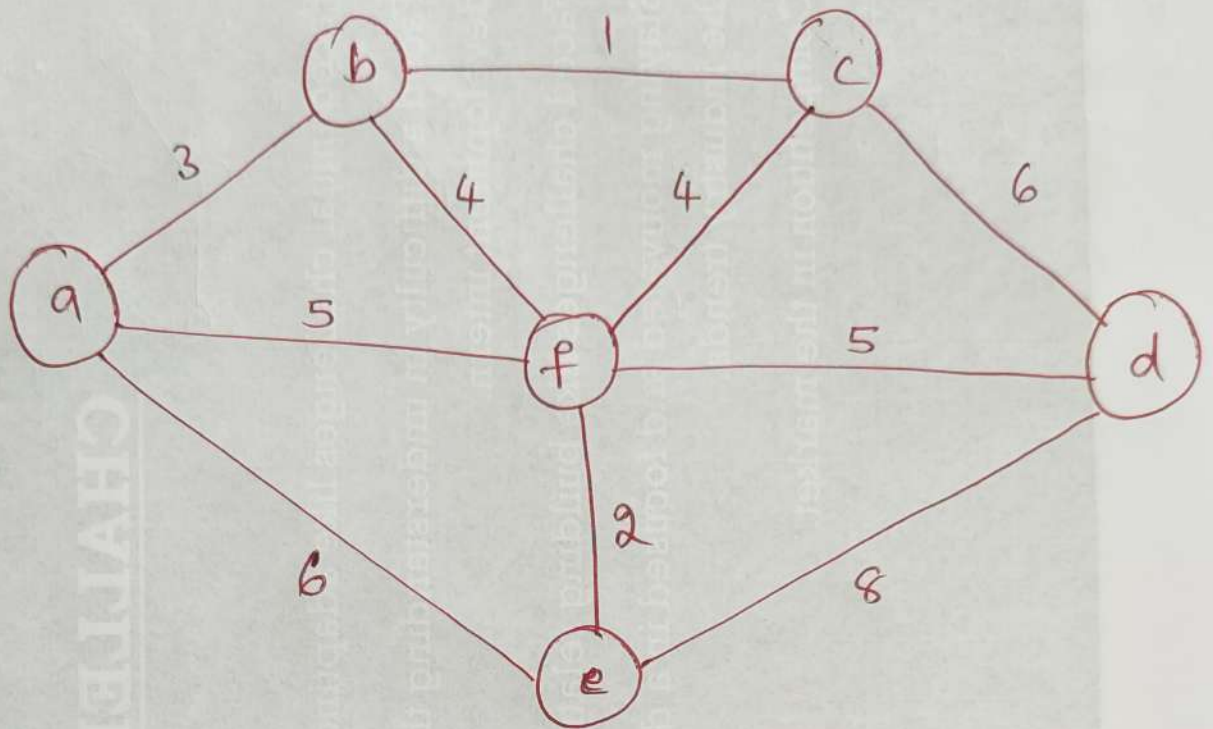


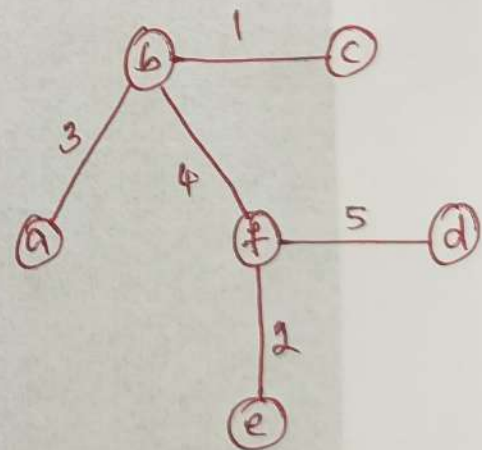
① Krushkal Algorithm / Prims Algorithm



Cost Matrix

	a	b	c	d	e	f
a	0	3	999	999	6	5
b	3	0	1	999	999	4
c	999	1	0	6	999	4
d	999	999	6	0	8	5
e	6	999	999	8	0	2
f	5	4	4	5	2	0

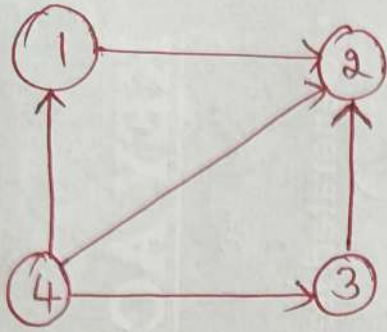
Output.



Minimum Cost = 15

TOPOLOGICAL ORDERING

Input 1:

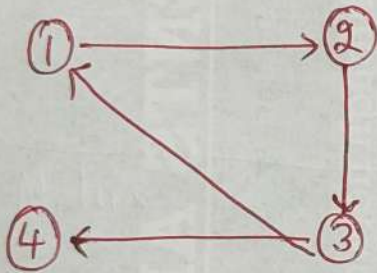


Adjacency Matrix

	①	②	③	④
①	0	1	0	0
②	0	0	0	0
③	0	1	0	0
④	1	1	1	0

Output :- 4 1 3 2

Input 2:



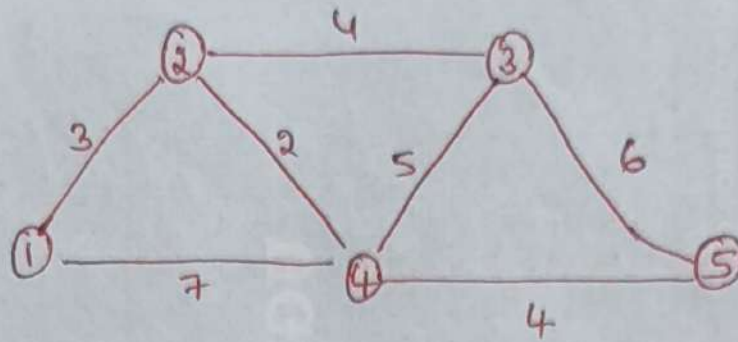
Adjacency Matrix

	①	②	③	④
①	0	1	0	0
②	0	0	1	0
③	1	0	0	1
④	0	0	0	0

Note:- Observe the arrow marks.

Output :- Topological Ordering
Not possible.

DIJKSTRA'S ALGORITHM.



Adjacency Matrix

	①	②	③	④	⑤
①	0	3	999	7	999
②	3	0	4	2	999
③	999	4	0	5	6
④	7	2	5	0	4
⑤	999	999	6	4	0

OUTPUT

$$a \rightarrow b = 3$$

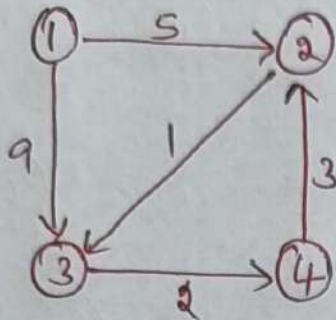
~~a → b → c = 7~~

$$a \rightarrow b \rightarrow d = 5$$

$$a \rightarrow b \rightarrow c = 7$$

$$a \rightarrow b \rightarrow d \rightarrow e = 9.$$

FLOYDS ALGORITHM



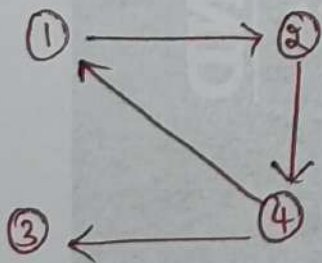
Adjacency Matrix

	①	②	③	④
①	0	5	9	999
②	999	0	1	999
③	999	999	0	2
④	999	3	999	0

Output

	①	②	③	④
①	0	5	6	8
②	999	0	1	3
③	999	5	0	2
④	999	3	4	0

WARSHALLS ALGORITHM



Adjacency Matrix

	①	②	③	④
①	0	1	0	0
②	0	0	0	1
③	0	0	0	0
④	1	0	1	0

OUTPUT

	①	②	③	④
①	1	1	1	1
②	1	1	1	1
③	0	0	0	0
④	1	1	1	1